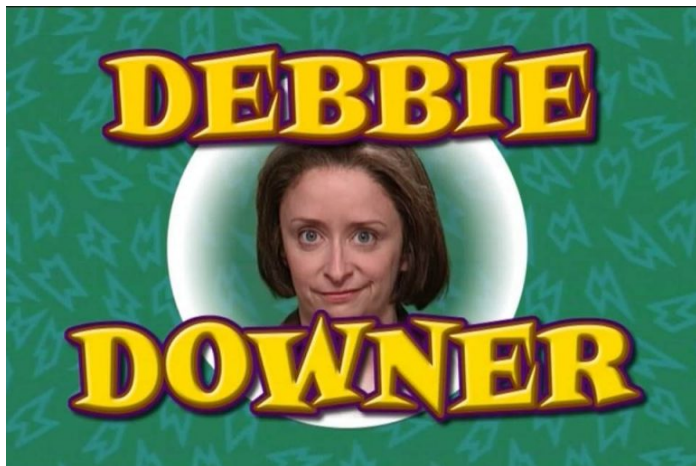


Regression assumptions and data cleaning

David Puelz

The University of Austin



Data cleaning

Regression assumptions

Influential observations

Multicollinearity

“90% of statistics is data cleaning.”

— Name missing



- Look for cases with out-of-range values that may indicate errors in the data



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases
- Look for cases with missing data



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases
- Look for cases with missing data
 - Can simply omit cases when data is missing in the variable(s) you plan to use—but then you lose all data in the other variables too!



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases
- Look for cases with missing data
 - Can simply omit cases when data is missing in the variable(s) you plan to use—but then you lose all data in the other variables too!
 - Can try to impute values for missing data by running a regression to predict the missing values from the other variables



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases
- Look for cases with missing data
 - Can simply omit cases when data is missing in the variable(s) you plan to use—but then you lose all data in the other variables too!
 - Can try to impute values for missing data by running a regression to predict the missing values from the other variables
- If cases are not Missing Completely At Random (MCAR), then omitting cases with missing data may bias your conclusions



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases
- Look for cases with missing data
 - Can simply omit cases when data is missing in the variable(s) you plan to use—but then you lose all data in the other variables too!
 - Can try to impute values for missing data by running a regression to predict the missing values from the other variables
- If cases are not Missing Completely At Random (MCAR), then omitting cases with missing data may bias your conclusions
- For categorical data with missing variables, can simply recode missing values as a special “unknown” category



- Look for cases with out-of-range values that may indicate errors in the data
 - When data is clearly in error and it's not clear how to fix it, usually best to just omit those cases
- Look for cases with missing data
 - Can simply omit cases when data is missing in the variable(s) you plan to use—but then you lose all data in the other variables too!
 - Can try to impute values for missing data by running a regression to predict the missing values from the other variables
- If cases are not Missing Completely At Random (MCAR), then omitting cases with missing data may bias your conclusions
- For categorical data with missing variables, can simply recode missing values as a special “unknown” category
- No easy solutions!

We'll use the college data set to predict a school's graduation rate from various factors. Take a look at the data; what potential issues do you see? We'll be using the following variables:

- Graduation rate
- Acceptance rate
- SAT score variables
- In-state tuition
- Out-of-state tuition

Many colleges have no SAT scores reported, so let's ignore those colleges (to enable a fair comparison) and also remove colleges with an obviously incorrect graduation rate of $> 100\%$:

```
my.sample <- colleges %>%  
  filter(!is.na(Average.combined.SAT) &  
         Graduation.rate <= 100)
```

Data cleaning

Regression assumptions

Influential observations

Multicollinearity

```

model <- lm(Graduation.rate ~ Average.combined.SAT + In.state.tuition,
            data=my.sample)
summary(model)

Call:
lm(formula = Graduation.rate ~ Average.combined.SAT + In.state.tuition,
    data = my.sample)

Residuals:
    Min       1Q   Median       3Q      Max
-45.526  -9.182   0.051   8.704  43.661

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -8.3246456   4.3708279   -1.905   0.0572 .
Average.combined.SAT  0.0611221  0.0048878  12.505   <2e-16 ***
In.state.tuition     0.0012486  0.0001111  11.237   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.75 on 709 degrees of freedom
(19 observations deleted due to missingness)
Multiple R-squared:  0.4469, Adjusted R-squared:  0.4453
F-statistic: 286.4 on 2 and 709 DF,  p-value: < 2.2e-16

```



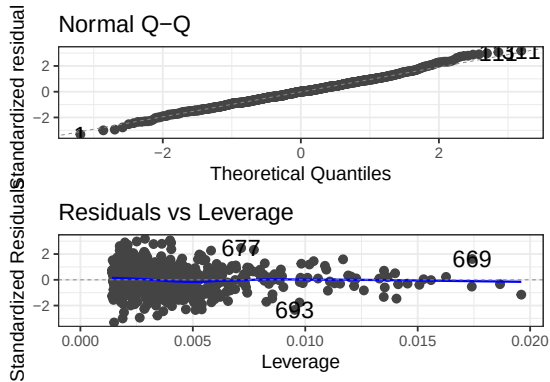
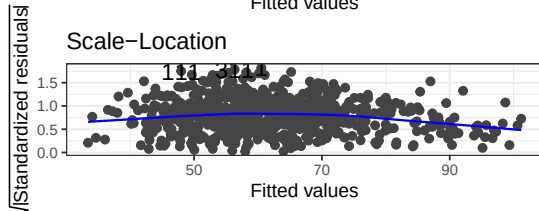
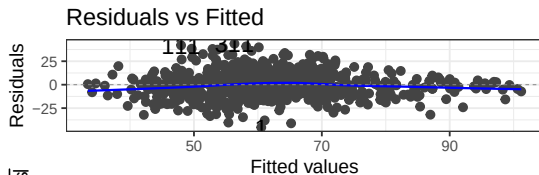

We need four things to be true for statistical inference (i.e., hypothesis tests, p -values, confidence intervals) to work for multiple regression:



We need four things to be true for statistical inference (i.e., hypothesis tests, p -values, confidence intervals) to work for multiple regression:

1. The errors are independent.
2. Y is a linear function of the X 's (except for the errors).
3. The errors are normally distributed.
4. The variance of Y is the same for any value of X ("homoscedasticity").

```
library(ggfortify)
autoplot(model) + theme_bw(base_size=9)
```





- Independence means that knowing the error (over-/under-prediction by the regression line) for one case doesn't tell you anything about the error for another case



- Independence means that knowing the error (over-/under-prediction by the regression line) for one case doesn't tell you anything about the error for another case
- Independence is usually most problematic with time series data (i.e., $X = \text{time}$)



- Independence means that knowing the error (over-/under-prediction by the regression line) for one case doesn't tell you anything about the error for another case
- Independence is usually most problematic with time series data (i.e., $X = \text{time}$)
- Since each college is completely separate, there is no reason to think the errors are not independent



- Independence means that knowing the error (over-/under-prediction by the regression line) for one case doesn't tell you anything about the error for another case
- Independence is usually most problematic with time series data (i.e., $X = \text{time}$)
- Since each college is completely separate, there is no reason to think the errors are not independent
- However, we could have a violation of independence if e.g. all of the colleges in Texas (say) all implemented the same policies to try to improve graduation rates

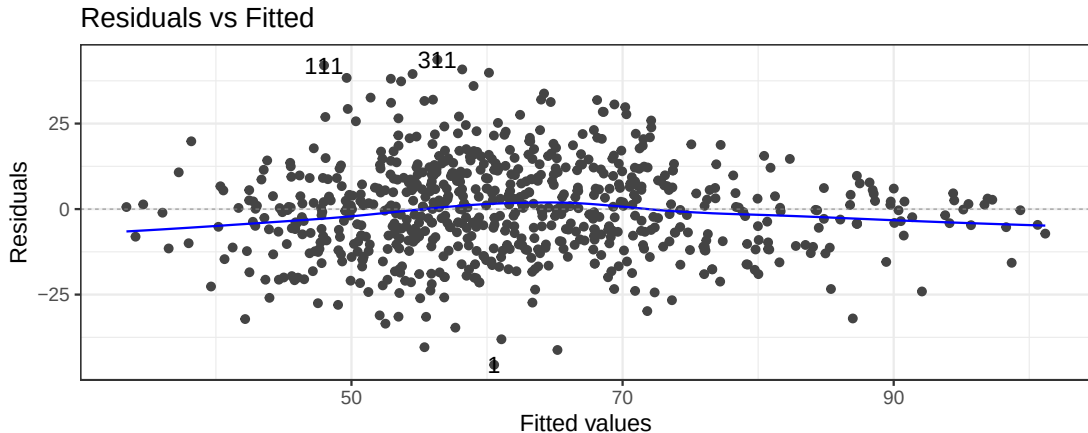


1. The errors are independent. ✓
2. Y is a linear function of the X 's (except for the errors).
3. The errors are normally distributed.
4. The variance of Y is the same for any value of X ("homoscedasticity").

Assumption 2: Linearity



Look at the residual plot—there should be **no trend** (the blue line should be roughly horizontal):



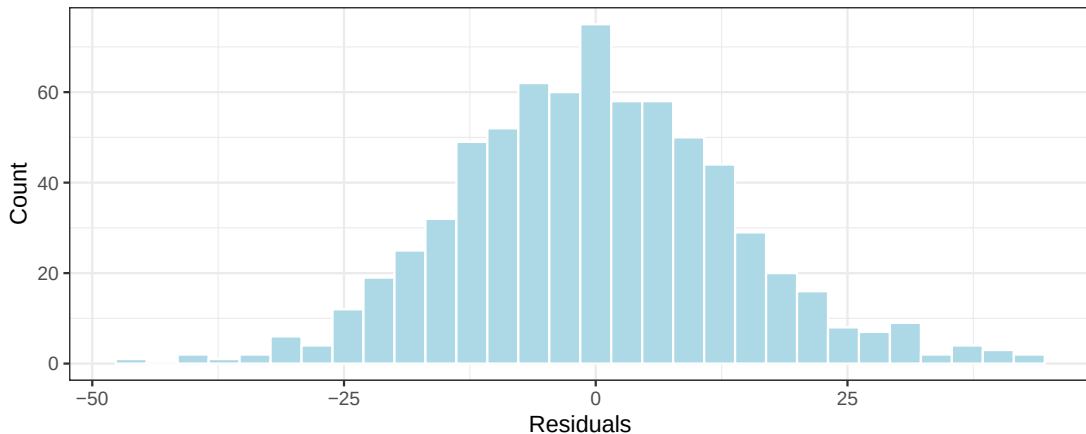


1. The errors are independent. ✓
2. Y is a linear function of the X 's (except for the errors). ✓
3. The errors are normally distributed.
4. The variance of Y is the same for any value of X ("homoscedasticity").

Assumption 3: Normality of residuals



We need the histograms to be Normally distributed:

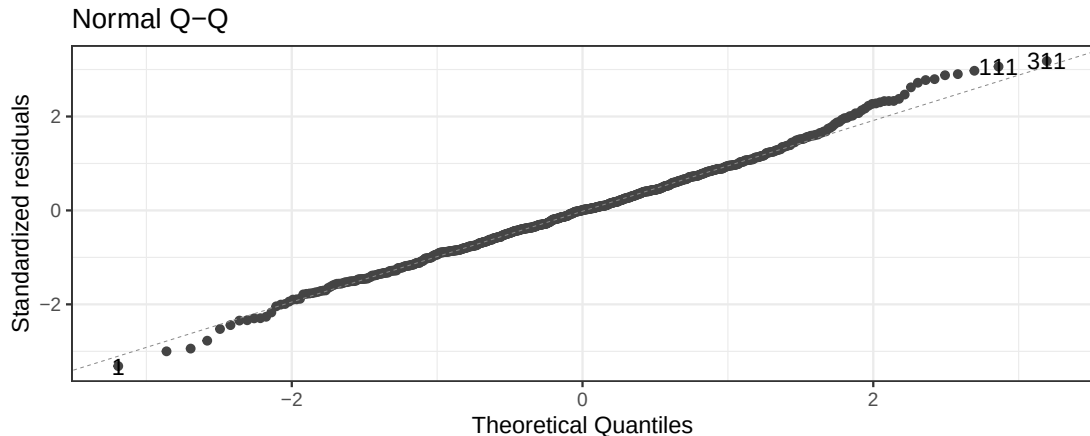


But it's hard to tell from a histogram!

Assumption 3: Normality of residuals



To be more careful we can use a Q-Q plot (a straight line indicates normality):



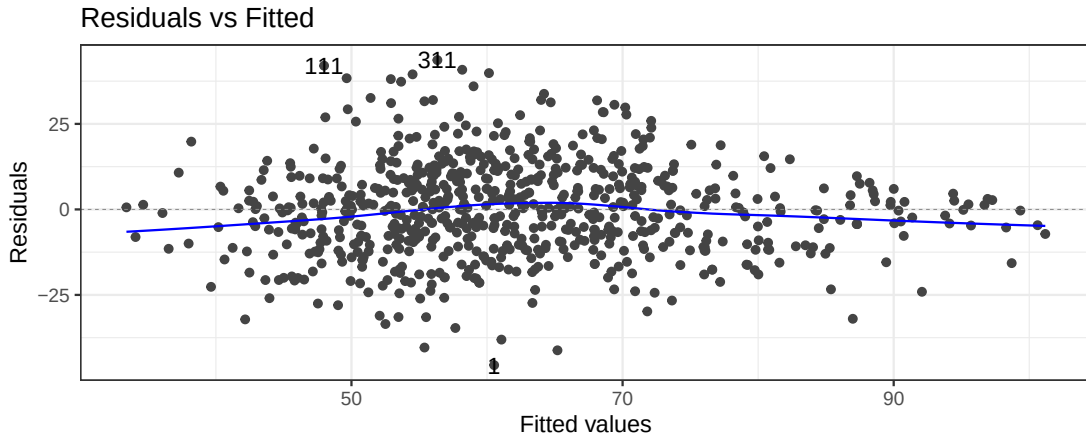


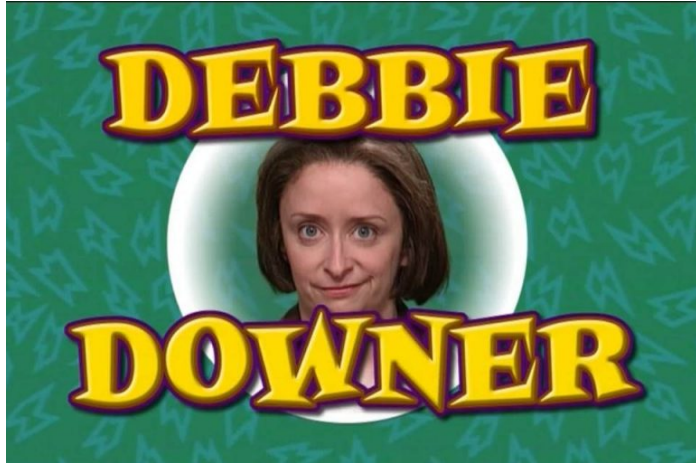
1. The errors are independent. ✓
2. Y is a linear function of the X 's (except for the errors). ✓
3. The errors are normally distributed. ✓
4. The variance of Y is the same for any value of X ("homoscedasticity").

Assumption 4: Homoscedasticity



Look at the residual plot—we want a roughly constant vertical “thickness”:







1. The errors are independent. ✓
2. Y is a linear function of the X 's (except for the errors). ✓
3. The errors are normally distributed. ✓
4. The variance of Y is the same for any value of X ("homoscedasticity"). ✗



- If any assumption is not satisfied, we should not trust the p -values or confidence intervals that come out of the model



- If any assumption is not satisfied, we should not trust the p -values or confidence intervals that come out of the model
- However, as long as linearity is satisfied, we can still use the model for making predictions (we just can't put reliable CIs on those predictions)



- If any assumption is not satisfied, we should not trust the p -values or confidence intervals that come out of the model
- However, as long as linearity is satisfied, we can still use the model for making predictions (we just can't put reliable CIs on those predictions)
- Is there anything else we can do? (Yes—stay tuned for next week!)

Data cleaning

Regression assumptions

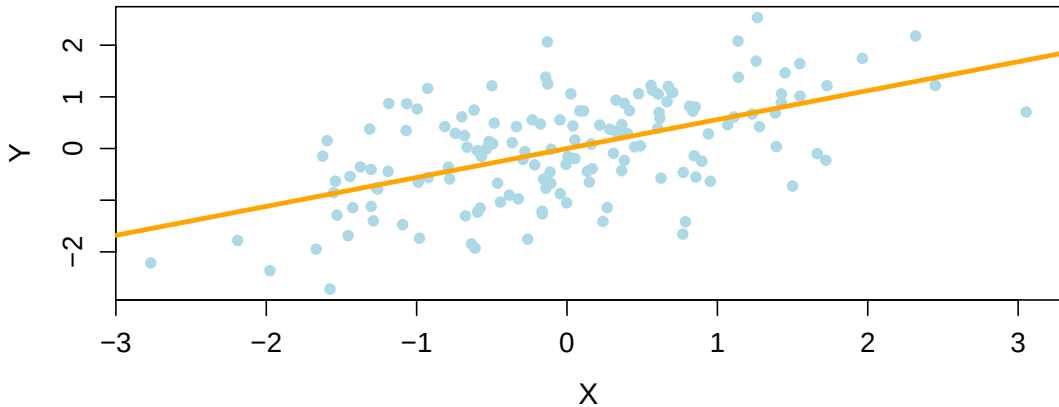
Influential observations

Multicollinearity

What a single case can do



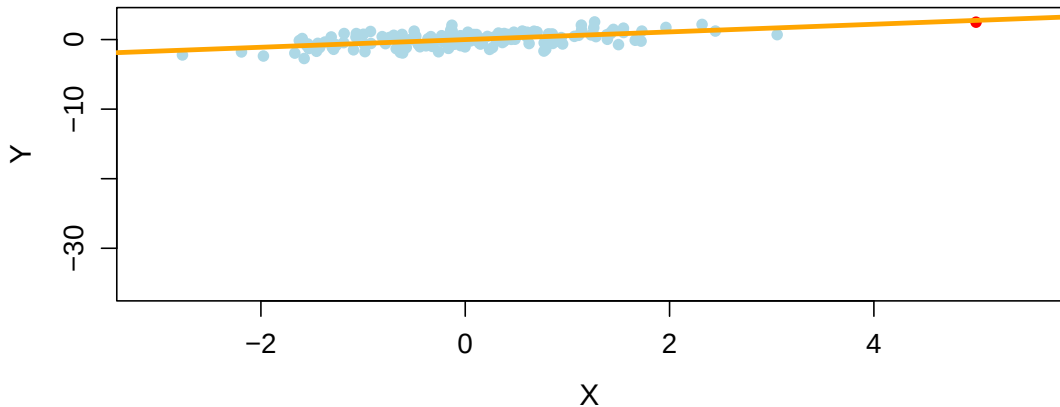
Let's take some hypothetical sample data:



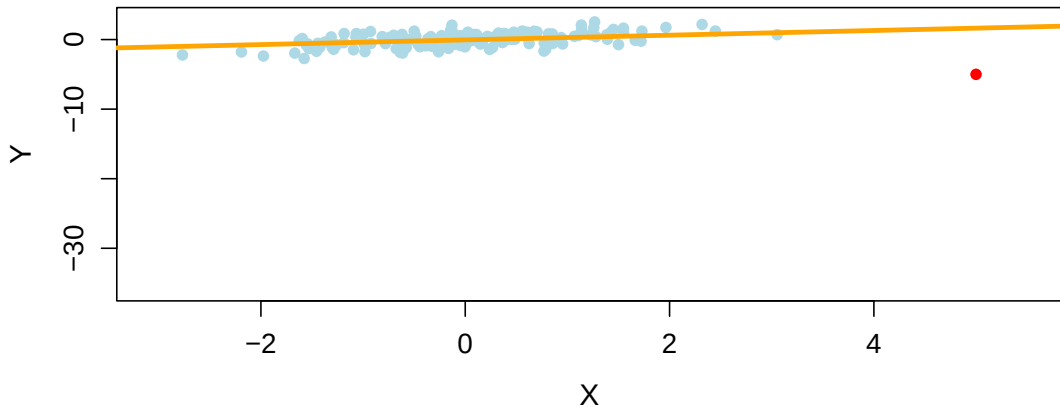


Even a single case can wreak havoc on the regression line. Let's add one outlier, at $X = 5$, and see what happens with different Y values.

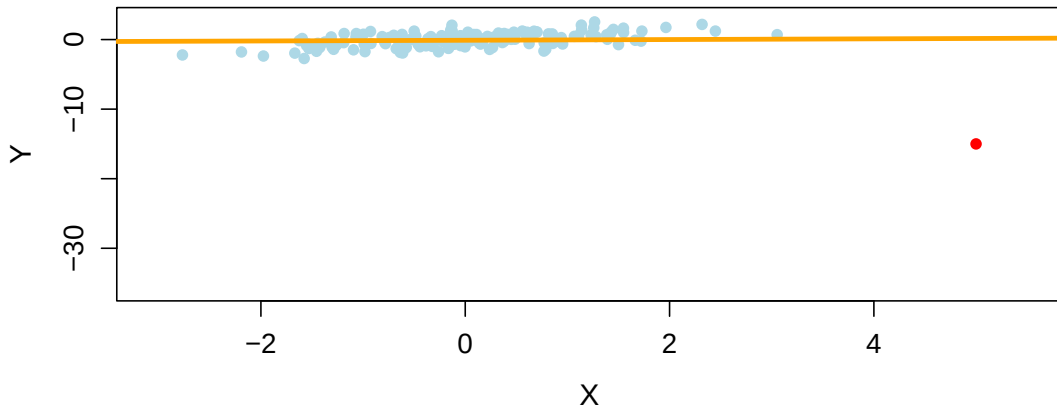
What a single case can do



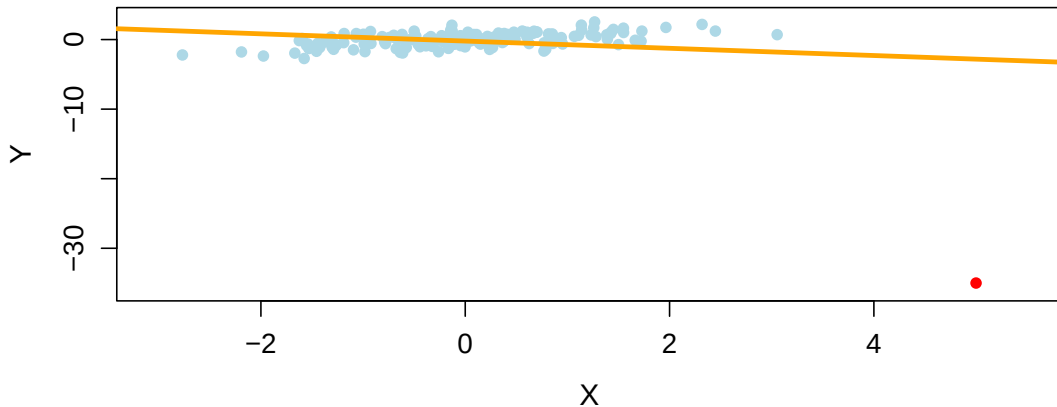
What a single case can do



What a single case can do



What a single case can do







Blackmail:

- Compromising information gives a blackmailer **leverage**—the *potential* to have a big impact
- Once the blackmailer uses the information, that gives them **influence**



Blackmail:

- Compromising information gives a blackmailer **leverage**—the *potential* to have a big impact
- Once the blackmailer uses the information, that gives them **influence**

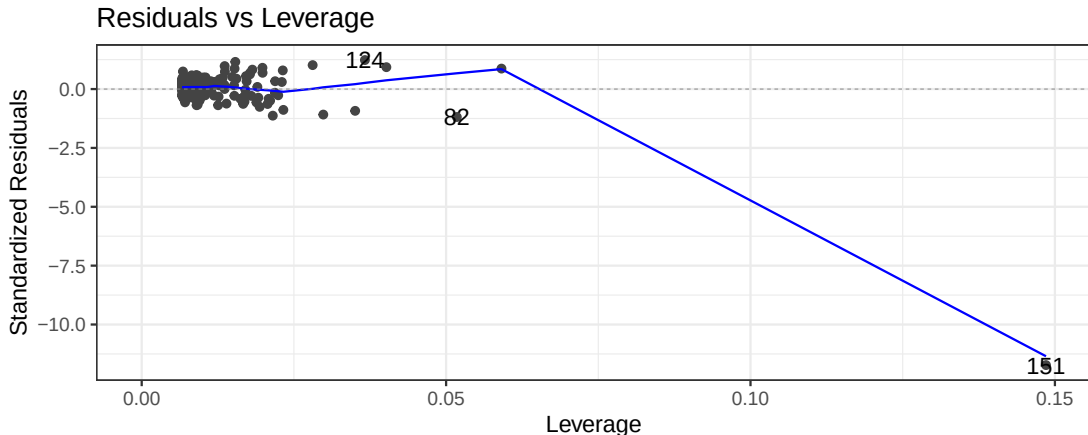
Regression:

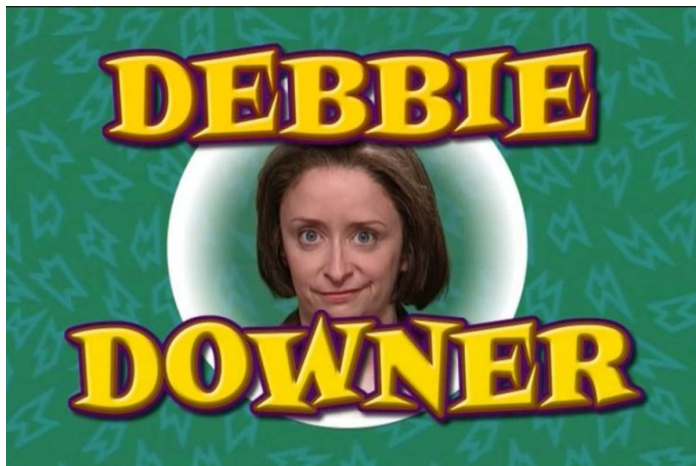
- When a case has a very unusual X value (i.e., far from $\overline{X}$), it has **leverage**—the *potential* to have a big impact on the regression line
- When that case *also* has a Y value that is out of line with the general trend, it will pull the regression line towards it—giving it **influence**

How do I know what cases are influential?



Look for cases with high leverage and a large (positive or negative) residual:







- If removing the influential observation doesn't make a substantial difference in your analysis, don't worry about it



- If removing the influential observation doesn't make a substantial difference in your analysis, don't worry about it
- If it does:



- If removing the influential observation doesn't make a substantial difference in your analysis, don't worry about it
- If it does:
 - Consider whether it could be a mistake (happens more than you might think!); if it is, correct the error or drop the case



- If removing the influential observation doesn't make a substantial difference in your analysis, don't worry about it
- If it does:
 - Consider whether it could be a mistake (happens more than you might think!); if it is, correct the error or drop the case
 - If not, hold out the influential observation(s) and report on them separately



- If removing the influential observation doesn't make a substantial difference in your analysis, don't worry about it
- If it does:
 - Consider whether it could be a mistake (happens more than you might think!); if it is, correct the error or drop the case
 - If not, hold out the influential observation(s) and report on them separately
 - Do not just throw out and ignore influential observations!

Data cleaning

Regression assumptions

Influential observations

Multicollinearity



Multicollinearity exists whenever 2+ predictors in a regression model are moderately or highly correlated.



Multicollinearity exists whenever 2+ predictors in a regression model are moderately or highly correlated.

- If two predictors X_1 and X_2 are highly correlated, it is hard to estimate the effect of changing X_1 while keeping X_2 constant.



Multicollinearity exists whenever 2+ predictors in a regression model are moderately or highly correlated.

- If two predictors X_1 and X_2 are highly correlated, it is hard to estimate the effect of changing X_1 while keeping X_2 constant.
- This means we will have large standard errors, and large p-values, for X_1 and/or X_2 .



Multicollinearity exists whenever 2+ predictors in a regression model are moderately or highly correlated.

- If two predictors X_1 and X_2 are highly correlated, it is hard to estimate the effect of changing X_1 while keeping X_2 constant.
- This means we will have large standard errors, and large p-values, for X_1 and/or X_2 .
- This **does not** mean there isn't a relationship between X_1 and Y , or X_2 and Y – it just means we can't pin down that relationship, because of the correlation!



Multicollinearity exists whenever 2+ predictors in a regression model are moderately or highly correlated.

- If two predictors X_1 and X_2 are highly correlated, it is hard to estimate the effect of changing X_1 while keeping X_2 constant.
- This means we will have large standard errors, and large p-values, for X_1 and/or X_2 .
- This **does not** mean there isn't a relationship between X_1 and Y , or X_2 and Y – it just means we can't pin down that relationship, because of the correlation!



Multicollinearity exists whenever 2+ predictors in a regression model are moderately or highly correlated.

- If two predictors X_1 and X_2 are highly correlated, it is hard to estimate the effect of changing X_1 while keeping X_2 constant.
- This means we will have large standard errors, and large p-values, for X_1 and/or X_2 .
- This **does not** mean there isn't a relationship between X_1 and Y , or X_2 and Y – it just means we can't pin down that relationship, because of the correlation!

Correlation between the response and the predictors is good, but correlation between the predictors is not!



We want to avoid multicollinearity in our models (if we can)!



We want to avoid multicollinearity in our models (if we can)!

- Any conclusions based on the p-values, coefficients, and confidence intervals of the highly correlated variables will be unreliable.



We want to avoid multicollinearity in our models (if we can)!

- Any conclusions based on the p-values, coefficients, and confidence intervals of the highly correlated variables will be unreliable.
- These statistics will not be stable: adding new data or predictors to the model could drastically change them.

How can we detect multicollinearity?



One way to see if two variables are collinear is to check the correlation between the two:

```
cor(my.sample$Average.math.SAT,  
    my.sample$Average.verbal.SAT,  
    use="complete.obs")
```

```
[1] 0.9194207
```

Any large correlation is potentially problematic.

How can we detect multicollinearity?



One way to see if two variables are collinear is to check the correlation between the two:

```
cor(my.sample$Average.math.SAT,  
    my.sample$Average.verbal.SAT,  
    use="complete.obs")
```

```
[1] 0.9194207
```

Any large correlation is potentially problematic. But what if there is multicollinearity among 3+ predictors?

A better way to check multicollinearity is using Variance Inflation Factors (VIF).

- The VIF is

$$\text{VIF}(\beta_j) = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 in a regression predicting X variable j from the other X variables.

A better way to check multicollinearity is using Variance Inflation Factors (VIF).

- The VIF is

$$\text{VIF}(\beta_j) = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 in a regression predicting X variable j from the other X variables.

- $\text{VIF}(\beta_j) = 0$ when $R_j^2 = 0$; i.e., the j th predictor variable is completely independent from the others.

A better way to check multicollinearity is using Variance Inflation Factors (VIF).

- The VIF is

$$\text{VIF}(\beta_j) = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 in a regression predicting X variable j from the other X variables.

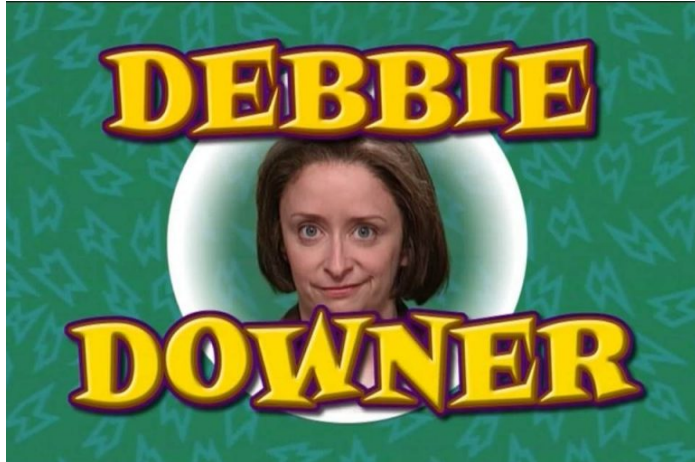
- $\text{VIF}(\beta_j) = 0$ when $R_j^2 = 0$; i.e., the j th predictor variable is completely independent from the others.
- $\text{VIF}(\beta_j)$ increases as R_j^2 does, and is ∞ when there is perfect multicollinearity; i.e., when X_j is perfectly predictable from the other X variables.

To calculate VIF for each predictor, you would have to run one regression for each predictor. But you don't have to run all of these regressions by hand!

```
model <- lm(Graduation.rate ~ Average.math.SAT +  
            Average.verbal.SAT + Acceptance.rate,  
            data=my.sample)  
library(car)  
vif(model)
```

Average.math.SAT	Average.verbal.SAT	Acceptance.rate
6.647073	6.454614	1.224408

Predictors with $VIF > 5$ are a cause for concern.





There are two general strategies for dealing with multicollinearity:

- Drop one of the variables with a high VIF factor, and rerun to see if VIFs have improved. (Just like we drop one of the dummy variables when putting a categorical variable in the model!)

And a third option – Accept it!



There are two general strategies for dealing with multicollinearity:

- Drop one of the variables with a high VIF factor, and rerun to see if VIFs have improved. (Just like we drop one of the dummy variables when putting a categorical variable in the model!)
- Combine the variables that correlate into a composite variable. (Combined SAT score = Math + Verbal)

And a third option – Accept it!

```
summary(lm(Graduation.rate ~ Average.combined.SAT + Acceptance.rate,  
          data=my.sample))
```

Call:

```
lm(formula = Graduation.rate ~ Average.combined.SAT + Acceptance.rate,  
    data = my.sample)
```

Residuals:

Min	1Q	Median	3Q	Max
-46.962	-9.619	-0.668	8.877	47.632

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-27.632646	6.619264	-4.175	3.35e-05 ***
Average.combined.SAT	0.090296	0.004967	18.179	< 2e-16 ***
Acceptance.rate	0.024146	0.039224	0.616	0.538

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.91 on 725 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.3507, Adjusted R-squared: 0.3489

F-statistic: 195.8 on 2 and 725 DF, p-value: < 2.2e-16



When is collinearity *not* an issue?

- When there is high collinearity in X 's that are strictly for adjustment, not interpretation, and the coefficient we want to interpret corresponds to a variable with low multicollinearity
- When multicollinearity comes from how we construct X – e.g., adding polynomial terms (next week!), turning categories into dummy variables, or adding interactions.
- When we are just trying to predict using relatively few X 's, and only predicting at “typical” X values.